Lux Network
The Institutional Substrate
for Regulated Digital Finance

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2026

What Lux is

Lux is the institutional substrate for regulated digital finance. Three things, sold as one through W3A Foundation (`w3a.foundation`):

1. A production sovereign Layer-1 blockchain (Lux Network) with post-quantum consensus (Quasar: parallel BLS + Pulsar + Corona certificates), native compliance precompiles (ERC-3643 + OnchainID + ML-DSA / SLH-DSA / FHE), threshold MPC custody as a first-class primitive, and a registered-transfer-agent signing pattern baked into the protocol.

2. A foundational patent estate covering the operator-plus-validator-plus-execution-gate architecture that underlies every modern Layer-2 rollup, every cross-chain messaging protocol, every alternative-trading-system stack, every broker-dealer onboarding flow, every transfer-agent cap table, every fund administration platform, every banking on/off-ramp, every FHE-confidential compliance gate, and every DEX with compliance gating. Anchored by U.S. Patent 11,803,853 B2 (2023, acquired). Extended by a 15-application post-quantum patent family filed March 3, 2026.

3. A licensing program operated through W3A Foundation. Membership-based. No automatic licenses. Every commercial operator practicing the patented architecture requires W3A Foundation membership.

The combination is unique. There is no other entity that owns the foundational rollup IP, operates the licensed reference substrate, and provides the regulatory compliance stack as a complete commercial offering.

Why now

Three forces converge on a 24-to-60-month window.

Tokenization is going institutional. BlackRock USD Institutional Digital Liquidity Fund (BUIDL) crossed $1.7B AUM within twelve months of launch. Franklin Templeton BENJI, Ondo Finance OUSG and USDY, Hamilton Lane SCOPE, Apollo Diversified Credit, KKR Health Care Strategic Growth, and ten other named institutional tokenized vehicles are scaling now. Forrester, McKinsey, and BCG forecast $5–16T in tokenized assets by 2030. The infrastructure-spending base alone (the picks and shovels of tokenization) is projected at $50–200B annually post-2030.

Regulatory mandate is forcing post-quantum migration. NIST finalized FIPS 203/204/205 in 2024 and selected HQC in 2025. The Office of the Comptroller of the Currency, the Federal Reserve, the SEC, the European Central Bank, the German BSI, the U.K. NCSC, and the Monetary Authority of Singapore have all issued post-quantum migration guidance for financial-sector critical infrastructure. The migration window is 2027–2032. Every institutional digital-finance platform will need to migrate. The Lux post-quantum patent family covers every migration path.

IP is no longer a free option. Every modern Layer-2 rollup, every cross-chain messaging protocol, every alt-DA architecture, and every multi-party tokenization stack reads on the foundational Nahmii architecture acquired by Lux Industries in 2025. The acquisition closes the window in which IP was a non-issue for the rollup-and-tokenization industry. Lux's enforcement program begins selectively in 2026, with Tier-1 settlements paving the way for Tier-3 escalation against incumbents.

1 The Lux substrate

1.1 Architecture

A short summary of the technology layer. Detailed in separate documents.

- **Quasar consensus.** Post-quantum Layer-1 consensus with parallel BLS + Pulsar + Corona certificates. Configurable profiles support classical-only, parallel-classical-plus-PQ, hybrid, strict-PQ, and maximum-strict modes.
- **Lux EVM.** EVM-compatible execution layer with native precompiles for ML-DSA, SLH-DSA, FN-DSA, hybrid PQ+classical, FROST/CGGMP21 threshold signatures, BLS aggregate, Ed25519, secp256r1, FHE primitives, and the Lux atomic-settlement gateway.
- **Native compliance.** ERC-3643 + OnchainID identity registries; trusted-issuer registries; ResaleQuadrant, CrowdfundFirstYear, OffshoreDistributionPeriod compliance modules; transfer-agent signing authority as a protocol-level primitive.
- **Threshold MPC custody.** CGGMP21 ECDSA and FROST EdDSA threshold signing with transfer-agent share as cryptographic constituent (not procedural prerequisite).
- **Cross-chain bridging.** Native Lux Bridge for institutional flows; license-protected interoperability with major cross-chain messaging protocols.
- **Operator-grade tooling.** Lux Operator Kubernetes CRD for declarative deployment; Lux KMS for HSM-backed secret management; Lux IAM for OAuth2/OIDC; Lux Gateway for JWT-validated request routing.

1.2 Lux Industries operating model

Lux Industries Inc. (Delaware) holds the patent estate, operates the Lux Network mainnet validators, and licenses the substrate. White-label commercial tenants run on the Lux Network as branded operating platforms. Each tenant has its own brand, customer base, regulatory licensing, and product surface.

Tenant	**Use case**
Liquidity.io	U.S.-regulated ATS / BD / TA / atomic-settlement tokenization platform (`liquidity.io`, `satschel.com`)
MLC Industries	Specialty tokenization (Maru Liquidity Capital)
VCC Industries	Variable Capital Company (Singapore) tokenization vehicle
Pars Industries	Persian / MENA institutional tokenization (`pars.network`)
Osage Industries	Sovereign-nation institutional digital infrastructure (Osage Nation)
Hanzo Industries	AI/agent-economy substrate consumer of Lux IP
Zoo Industries	Open-research and consumer-facing infrastructure consumer

No automatic licensing. Each tenant must independently join W3A Foundation as a member at Strategic or General tier covering the tenant's field of use to obtain a license to practice the Lux patent estate. Tenants operating without W3A Foundation membership are operating without a license and are exposed to Lux Industries' enforcement program. Tenants operate independently as Delaware corporations with separate brands, customers, and balance sheets, and pursue W3A membership on their own commercial timeline.

2 The Lux patent estate

2.1 Asset summary

Asset	Subject matter	Status
US 11,803,853 B2	Foundational rollup architecture: operator + validator + execution gate	Acquired (2025), expires 2039
2026-001 CIP	PQ master extension of 11,803,853	Filing March 3, 2026
2026-002	PQ data-availability attestation	Filing March 3, 2026
2026-003	PQ cross-chain message gating	Filing March 3, 2026
2026-004	PQ institutional tokenization settlement (general)	Filing March 3, 2026
2026-005	Cryptographic algorithm agility framework	Filing March 3, 2026
2026-006	PQ threshold MPC for regulated custody	Filing March 3, 2026
2026-007	PQ parallel-certificate L1 consensus	Filing March 3, 2026
2026-008	PQ ATS / exchange matching and settlement	Filing March 3, 2026
2026-009	PQ broker-dealer customer onboarding and funding	Filing March 3, 2026
2026-010	PQ transfer-agent cap table and corporate actions	Filing March 3, 2026
2026-011	PQ banking fiat on-ramp and off-ramp	Filing March 3, 2026
2026-012	PQ tokenized fund administration and NAV	Filing March 3, 2026
2026-013	PQ atomic T+0 securities settlement	Filing March 3, 2026
2026-014	PQ + FHE confidential compliance gate	Filing March 3, 2026
2026-015	PQ DEX / order-book and AMM with compliance gating	Filing March 3, 2026

The estate covers every commercially relevant surface in the regulated digital finance stack: L1 consensus, L2 rollups, cross-chain messaging, ATS matching, broker-dealer customer operations, transfer-agent cap tables, fiat on/off-ramps, fund administration, atomic settlement, custody, and the algorithm-agility framework that ties them all together for the post-quantum migration window. Through 2046+ depending on prosecution timing.

2.2 Why the estate is durable

Priority date precedence. The foundational patent's 4 May 2018 priority date predates every modern rollup in production. The PQ family's 2026 priority date predates the regulatorily-mandated PQ migration window (2027–2032).

Plasma-and-MVP-aware claim construction. The strongest pre-2018 prior art is Plasma (August 2017) and Minimal Viable Plasma (January 2018). Lux's claim construction strategy distinguishes via threshold formalization, completeness/correctness dichotomy, and the two-agent architecture. Continuation strategy provides backstop.

Primitive-agnostic and architecture-agnostic PQ family. The PQ family does not specify a single PQ primitive; it claims the integration of post-quantum or hybrid signature primitives into the broader transaction-control architecture. Every PQ migration path (ML-DSA, SLH-DSA, FN-DSA, hybrid, future NIST primitives) enters the patent zone.

10-year inventor track record. Lux Industries leadership has been building the underlying primitives since 2015 (publicly visible at `github.com/arcalabs`: `arcadia` desktop wallet from August 2015; `ethereum-cluster` from April 2018; the `st-contracts` Arca Security Token protocol from January 2019). The 10-year track record establishes inventor-of-record credibility and provides defensive prior-art cover against derivation claims.

3 The licensing program

3.1 Tier 1 — Lux ecosystem

Available to: white-label commercial tenants of Lux Network running their products on Lux substrate. Liquidity.io, MLC Industries, VCC Industries, Pars Industries, Osage Industries, Hanzo Industries, Zoo Industries are the current Tier-1 licensees.

- Perpetual, royalty-free, non-transferable license to practice the entire Lux patent estate within the licensee's field of use.
- Cross-license-back covenant on patents the licensee develops.
- No-challenge covenant; defensive-termination clause.
- Field of use defined by each tenant's commercial domain.
- Confidential and back-office; does not affect customer-facing branding.

3.2 Tier 2 — Lux-certified substrate

Available to: third-party institutional buyers running tokenized funds, regulated trading platforms, or compliant rollups on Lux substrate.

- Platform fee for substrate access (validator infrastructure, KMS, IAM, gateway, monitoring).
- Running royalty on substrate-attributable revenue (0.25% to 1.00% based on volume tier).
- Includes Lux-certified-rollup certification (technical audit + compliance attestation).
- Includes the patent license bundled into the substrate offering.
- Most-favored-licensee protection within named institutional sub-segments.

Target customers: institutional asset managers operating tokenized funds; regulated broker-dealers expanding into tokenization; tokenized real-world-asset platforms; central-bank digital currency operators; institutional crypto custodians.

3.3 Tier 3 — External operator

Available to: operators of competing protocols, rollups, or tokenization stacks who wish to license without joining the Lux substrate.

- Lump-sum signing fee in the $10M–$200M range, scaling with the operator's exposure and AUM.
- Running royalty (1.0% to 3.0%) on patent-attributable revenue.
- No substrate access; standalone patent license only.
- Field-of-use restrictions; no most-favored-licensee unless negotiated separately.
- No-challenge covenant; defensive termination.

Target customers: Securitize, Tokeny, Polymesh, ADDX, INX, Apex Securities Inc.; OP Stack operators with Alt-DA configurations; Arbitrum Orbit AnyTrust chains; cross-chain messaging operators (LayerZero Labs, Chainlink Labs, Axelar Network); RaaS providers (Alchemy Rollups, AltLayer, Caldera, Conduit, Gelato Rollups).

4 Why institutional buyers choose Lux

The institutional value proposition is concrete and immediate.

IP-clean infrastructure. Lux is the only substrate where the operator simultaneously owns the foundational rollup and tokenization IP. Building on Lux means building on infrastructure that is not subject to third-party patent assertion. This is unique in the market.

Native compliance. ERC-3643, OnchainID, transfer-agent signing authority, threshold MPC custody, ResaleQuadrant compliance modules, and per-jurisdiction regulatory enforcement are first-class protocol primitives. Compliance is not bolt-on; it is the platform.

Post-quantum readiness. Lux ships with parallel BLS + Pulsar + Corona certificates today, with configurable strict-PQ profiles and a crypto-agility framework. The substrate is ready for the regulatory PQ migration window from day one.

Atomic T+0 settlement. The Lux-native atomic mint-and-burn pair settlement primitive provides instantaneous T+0 settlement of tokenized securities trades with all-or-nothing atomicity. Operationally simpler than DTC + DTCC; cryptographically auditable; per-trade reversible only through transfer-agent-authorized reversal.

Operational maturity. Lux Network has been in production with white-label tenants since 2024. Liquidity.io operates 12,000+ markets across U.S. regulated securities and spot crypto on the Lux substrate today. Operational lessons have been learned; production controls (push-only ACH, name-matching, velocity caps, multi-processor failover, withdrawal hold windows) are battle-tested.

Coalition of substrate participants. Every Tier-1 ecosystem tenant cross-licenses into the Lux IP estate. The estate grows with the ecosystem. Tier-2 licensees benefit from the cross-license inventory and the no-challenge participant pool.

5 Competitive position

5.1 Versus public Ethereum + bolt-on compliance (Securitize, Tokeny, Polymesh, ADDX path)

Public-Ethereum-plus-compliance stacks are exposed to Lux IP enforcement (the 2023 Nahmii patent and the 2026 PQ family both read against the compliance-gate-plus-token-contract architecture they use). They have no licensed alternative substrate. The path forward for these operators is to license Lux IP (Tier 3) or to migrate to a Lux-licensed substrate (Tier 2).

5.2 Versus public Layer-2 rollups (Base, Arbitrum, Optimism)

Public L2 rollups are exposed to Lux IP enforcement (the 2023 Nahmii patent reads against the operator-and-fraud-proof architecture). Coinbase, Optimism Foundation, and Arbitrum Foundation have not yet engaged on licensing; Lux's enforcement program intends to escalate to these incumbents after Tier-1 settlements establish patent strength.

5.3 Versus institutional permissioned ledgers (R3 Corda, Hyperledger Fabric, IBM Blockchain)

Permissioned-ledger architectures lack the integrated compliance stack, the native post-quantum cryptography, and the public-network composability that institutional tokenization increasingly requires. They occupy the legacy-on-premise corner; Lux occupies the modern-public-infrastructure corner.

5.4 Versus other sovereign L1s (Solana, Avalanche, Sui, Aptos)

Other sovereign L1s lack the regulated-finance compliance stack (no native ERC-3643, no transfer-agent signing primitive, no atomic-mint-burn settlement, no fund-administration infrastructure). They are general-purpose L1s, not institutional-tokenization substrates. Lux is purpose-built for the use case.

6 Engagement model

6.1 For institutional buyers (Tier 2)

Step 1. Initial scoping meeting (one hour). Lux Industries leadership + buyer institutional digital-assets team. Scope: substrate access requirements, regulatory regime, tokenization roadmap.

Step 2. Substrate access plan (two weeks). Lux Industries provides a detailed proposal covering substrate fees, royalty structure, certification timeline, and target deployment milestones.

Step 3. Pilot deployment (60–90 days). Lux operates a pilot tokenized vehicle for the buyer in the buyer's chosen jurisdiction.

Step 4. Production rollout (six to twelve months). Full migration of buyer's tokenized fund operations to Lux substrate.

6.2 For incumbent rollup or tokenization operators (Tier 3)

Step 1. Confidentiality agreement and initial scoping (two weeks).

Step 2. Term sheet exchange (four weeks). Lux Industries proposes Tier-3 license structure including lump-sum, running-royalty, field-of-use definition, and no-challenge covenant. Operator responds with counter-terms.

Step 3. Diligence access (eight weeks). Operator reviews patent prosecution histories, claim charts, and prior-art memoranda.

Step 4. Definitive agreement (twelve weeks). Outside counsel drafts; both parties counter-mark.

Step 5. Signing (within 90 days of substantive engagement).

7 Contact

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Confidentiality